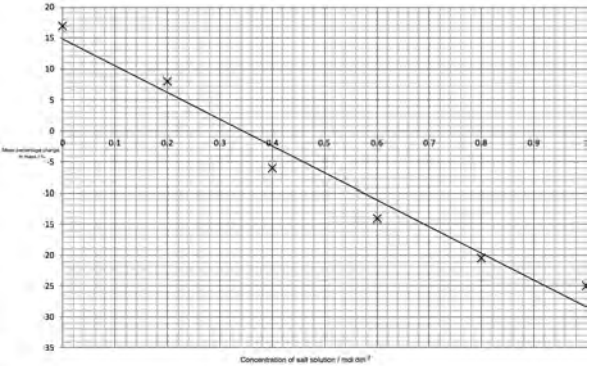


| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                            | Additional Guidance                                                                                                              | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1(a)</b>     | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• there is an intake of water (1)</li> <li>• (down a) water potential gradient outside to inside / (from) high WP to low WP / (from) low salt concentration (outside) and a high salt concentration (inside) (1)</li> <li>• by osmosis /described (1)</li> </ul> | <p>ACCEPT moves down a concentration gradient, in context of water entering.<br/>ACCEPT correct reference to hypo/hypertonic</p> | <b>(3)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Additional Guidance            | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------|
| <b>1(b)(i)</b>  | <p>A description that includes five of the following points:</p> <ul style="list-style-type: none"> <li>• cut pieces of same {size / surface area / potato} (1)</li> <li>• find mass of each piece before and after soaking (1)</li> <li>• soak pieces in solutions for {same length of time / stated time} (1)</li> <li>• blot dry before weighing (1)</li> <li>• repeat for each solution (1)</li> <li>• description of how percentage change calculated (1)</li> </ul> | <p>ACCEPT correct equation</p> | <b>(5)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                      | Additional Guidance                                                                                                                     | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1(b)(ii)</b> | <p>A graph showing the following features:</p> <ul style="list-style-type: none"> <li>A axes correct (x – concentration of sodium chloride, y - mean percentage change in mass) (1)</li> <li>L axes correctly labelled, and with units for x mol dm<sup>-3</sup> (1am4)</li> <li>P correct plotting on suitable linear scales (1)</li> <li>S suitable line of best fit drawn (1)</li> </ul> |  <p>S-NOT dot to dot, some points on either side</p> | <b>(4)</b> |

| Question Number  | Answer                                                                                                                                                                                                                                                                      | Additional Guidance                                                                                                                                | Mark       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1(b)(iii)</b> | <p>An answer showing the following stages:</p> <ul style="list-style-type: none"> <li>correct reading, from their graph in bii, of where line of best fit crosses 0% change in mass (1)</li> <li>use of this reading in biii graph to obtain water potential (1)</li> </ul> | <p>ecf applies</p> <p>e.g. 0.35 (mol dm<sup>-3</sup>)</p> <p>e.g. -1.7 (MPa)<br/>allow 0.1 MPa either side for any mol dm<sup>-3</sup> reading</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                            | Additional Guidance                                                                                                                                             | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1(c)</b>     | <p>An explanation that makes reference to three of the following points:</p> <ul style="list-style-type: none"> <li>the water potential falls initially, because {water is being lost / potatoes dry out} (1)</li> <li>fall in water potential levels off later because no longer a wp gradient / wp inside and out has reached equilibrium (1)</li> <li>water loss causes solute concentration in potato tissue cells to increase (1)</li> </ul> | <p>ACCEPT reference to up to week 11<br/>ACCEPT evaporates from potato</p> <p>ACCEPT reference to week 7 or beyond</p> <p>ACCEPT solute potential decreases</p> | <b>(3)</b> |